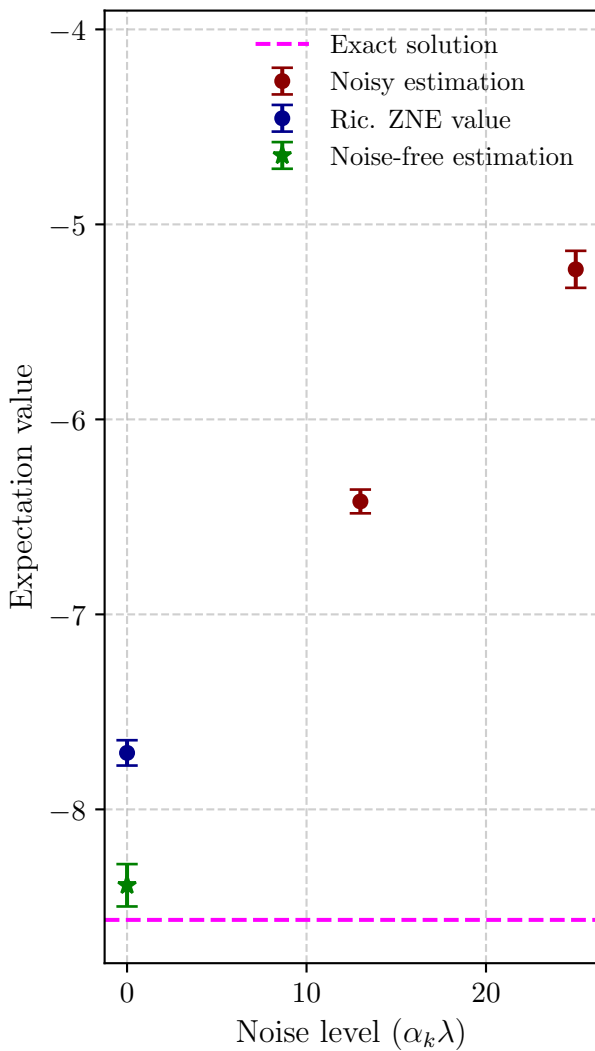
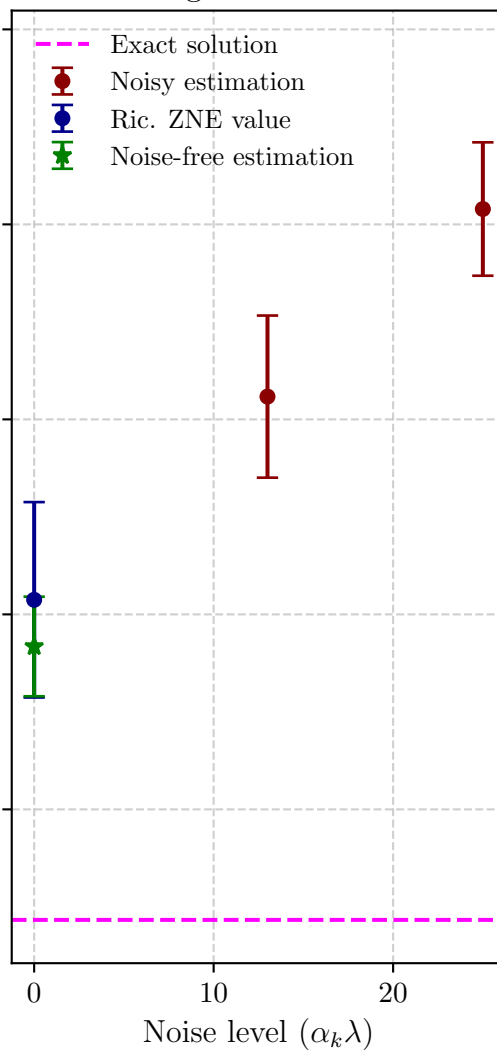


Description	Simulation	Mean $\pm$ Std
xy-ric2	vqe	-7.56107 $\pm$ 0.06520
xy-ric2	redundant (noise=13)	-6.42081 $\pm$ 0.06093
xy-ric2	redundant (noise=25)	-5.23054 $\pm$ 0.09492
xy-ric2	zne	-7.71027 $\pm$ 0.06494
xy-ric2	noiseoff	-8.38929 $\pm$ 0.10862
ising-ric2	vqe	-6.51507 $\pm$ 0.46899
ising-ric2	redundant (noise=13)	-5.88316 $\pm$ 0.41569
ising-ric2	redundant (noise=25)	-4.92123 $\pm$ 0.34205
ising-ric2	zne	-6.92525 $\pm$ 0.50107
ising-ric2	noiseoff	-7.16465 $\pm$ 0.25530
heisenberg-ric2	vqe	-6.26219 $\pm$ 0.17777
heisenberg-ric2	redundant (noise=13)	-5.54950 $\pm$ 0.19542
heisenberg-ric2	redundant (noise=25)	-4.35429 $\pm$ 0.22865
heisenberg-ric2	zne	-6.84432 $\pm$ 0.18240
heisenberg-ric2	noiseoff	-6.72634 $\pm$ 0.32239
xy-ric3-uniform-scaling	vqe	-7.56107 $\pm$ 0.06520
xy-ric3-uniform-scaling	redundant (noise=13)	-6.42081 $\pm$ 0.06093
xy-ric3-uniform-scaling	redundant (noise=25)	-5.23054 $\pm$ 0.09492
xy-ric3-uniform-scaling	redundant (noise=37)	-4.29102 $\pm$ 0.12477
xy-ric3-uniform-scaling	zne	-7.99323 $\pm$ 0.07628
xy-ric3-uniform-scaling	noiseoff	-8.38929 $\pm$ 0.10862
ising-ric3-uniform-scaling	vqe	-6.51507 $\pm$ 0.46899
ising-ric3-uniform-scaling	redundant (noise=13)	-5.88316 $\pm$ 0.41569
ising-ric3-uniform-scaling	redundant (noise=25)	-4.92123 $\pm$ 0.34205
ising-ric3-uniform-scaling	redundant (noise=37)	-4.12086 $\pm$ 0.29278
ising-ric3-uniform-scaling	zne	-7.10756 $\pm$ 0.52590
ising-ric3-uniform-scaling	noiseoff	-7.16465 $\pm$ 0.25530
heisenberg-ric3-uniform-scaling	vqe	-6.26219 $\pm$ 0.17777
heisenberg-ric3-uniform-scaling	redundant (noise=13)	-5.54950 $\pm$ 0.19542
heisenberg-ric3-uniform-scaling	redundant (noise=25)	-4.35429 $\pm$ 0.22865
heisenberg-ric3-uniform-scaling	redundant (noise=37)	-3.40588 $\pm$ 0.25160
heisenberg-ric3-uniform-scaling	zne	-7.12284 $\pm$ 0.17771
heisenberg-ric3-uniform-scaling	noiseoff	-6.72634 $\pm$ 0.32239
xy-ric4-uniform-scaling	vqe	-7.56107 $\pm$ 0.06520
xy-ric4-uniform-scaling	redundant (noise=13)	-6.42081 $\pm$ 0.06093
xy-ric4-uniform-scaling	redundant (noise=25)	-5.23054 $\pm$ 0.09492
xy-ric4-uniform-scaling	redundant (noise=37)	-4.29102 $\pm$ 0.12477
xy-ric4-uniform-scaling	redundant (noise=49)	-3.54359 $\pm$ 0.14586
xy-ric4-uniform-scaling	zne	-8.06125 $\pm$ 0.08022
xy-ric4-uniform-scaling	noiseoff	-8.38929 $\pm$ 0.10862
ising-ric4-uniform-scaling	vqe	-6.51507 $\pm$ 0.46899
ising-ric4-uniform-scaling	redundant (noise=13)	-5.88316 $\pm$ 0.41569
ising-ric4-uniform-scaling	redundant (noise=25)	-4.92123 $\pm$ 0.34205
ising-ric4-uniform-scaling	redundant (noise=37)	-4.12086 $\pm$ 0.29278
ising-ric4-uniform-scaling	redundant (noise=49)	-3.45033 $\pm$ 0.26011
ising-ric4-uniform-scaling	zne	-7.14435 $\pm$ 0.53243
ising-ric4-uniform-scaling	noiseoff	-7.16465 $\pm$ 0.25530
heisenberg-ric4-uniform-scaling	vqe	-6.26219 $\pm$ 0.17777
heisenberg-ric4-uniform-scaling	redundant (noise=13)	-5.54950 $\pm$ 0.19542
heisenberg-ric4-uniform-scaling	redundant (noise=25)	-4.35429 $\pm$ 0.22865
heisenberg-ric4-uniform-scaling	redundant (noise=37)	-3.40588 $\pm$ 0.25160
heisenberg-ric4-uniform-scaling	redundant (noise=49)	-2.64631 $\pm$ 0.26341
heisenberg-ric4-uniform-scaling	zne	-7.19007 $\pm$ 0.17634
heisenberg-ric4-uniform-scaling	noiseoff	-6.72634 $\pm$ 0.32239

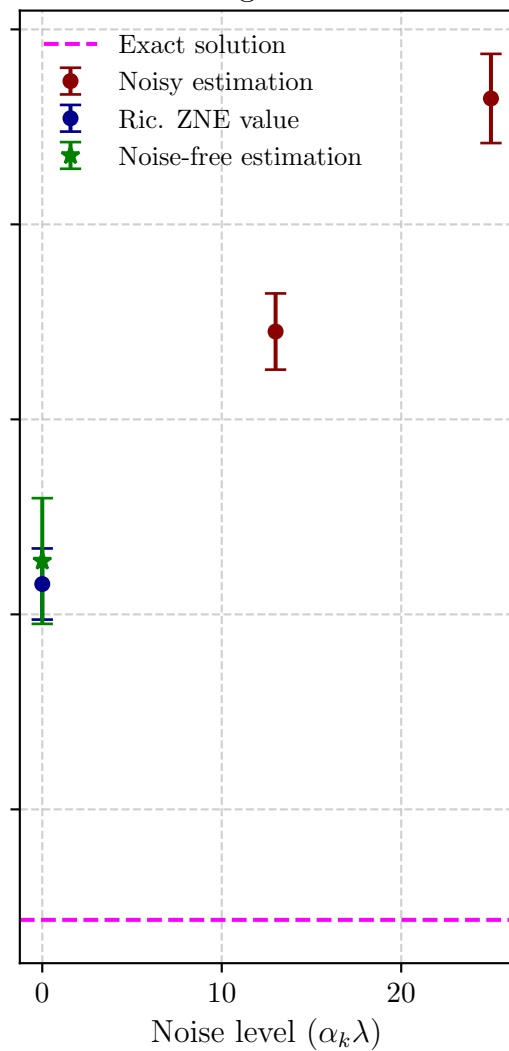
XY ansatz ric-2



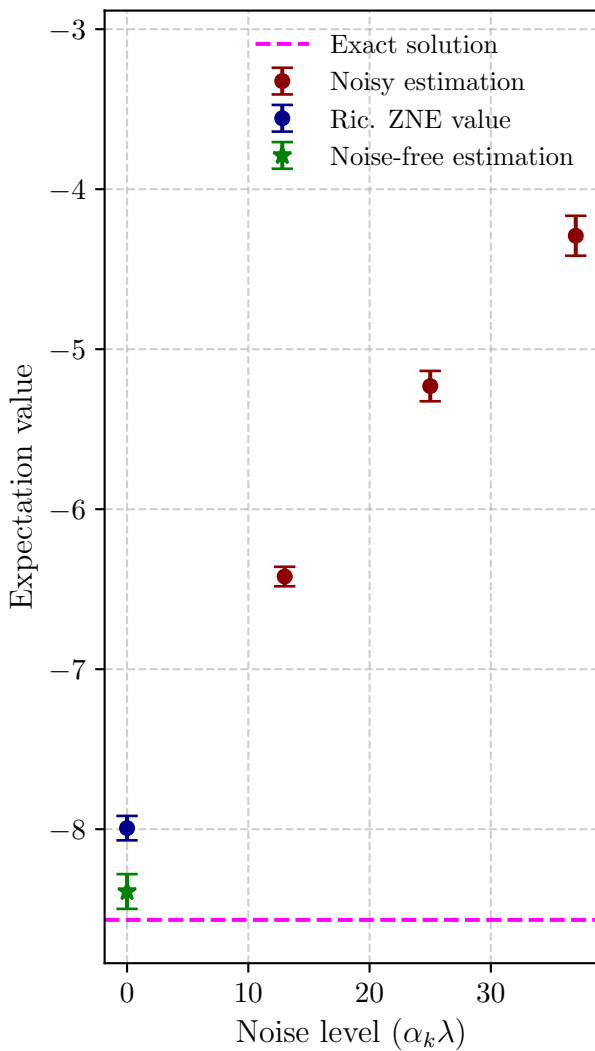
Ising ansatz ric-2



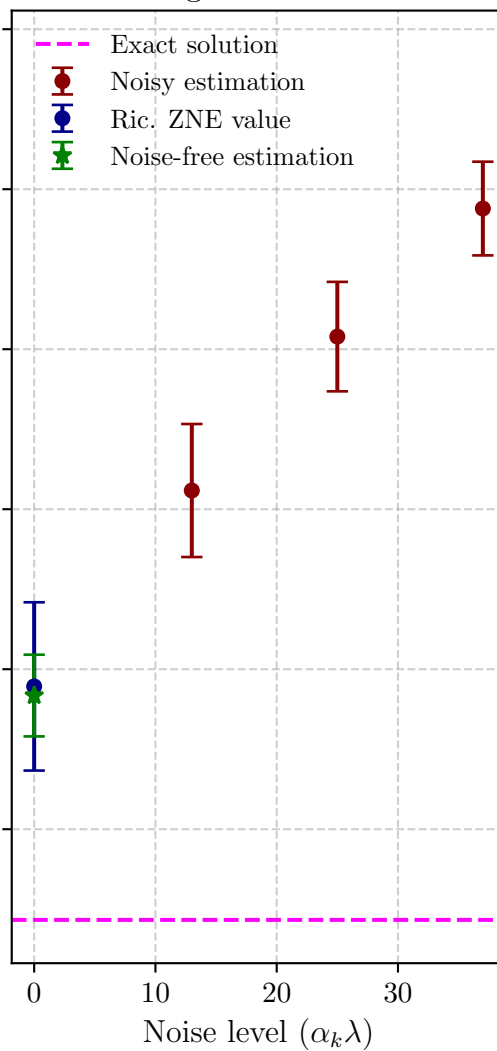
Heisenberg ansatz ric-2



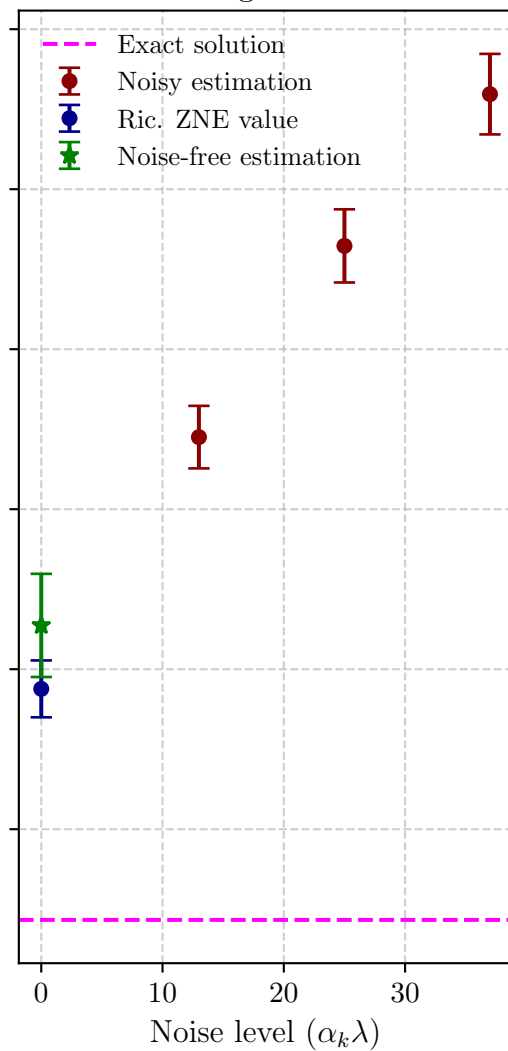
XY ansatz ric-3



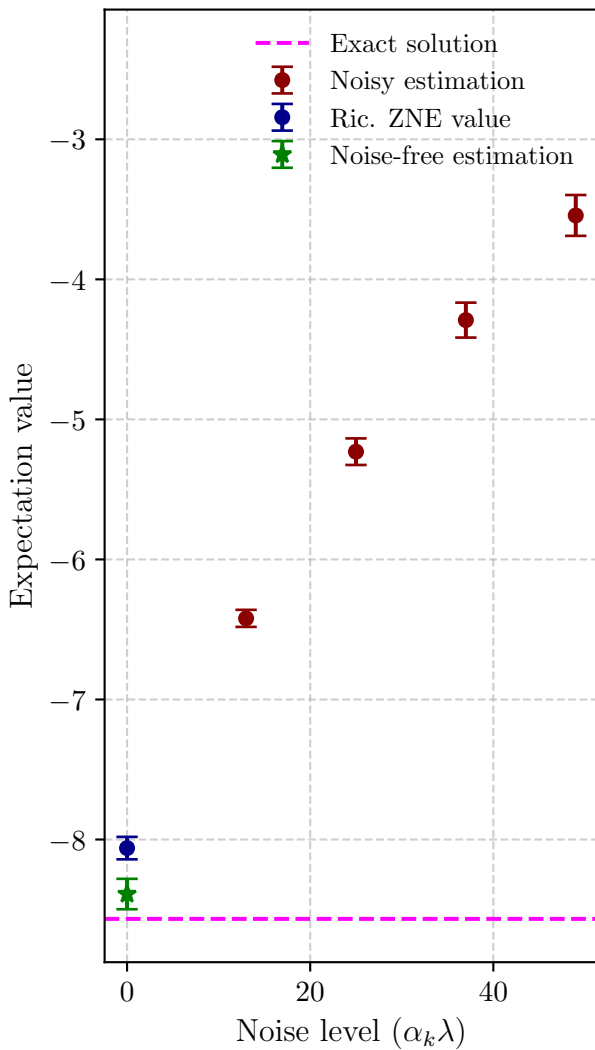
Ising ansatz ric-3



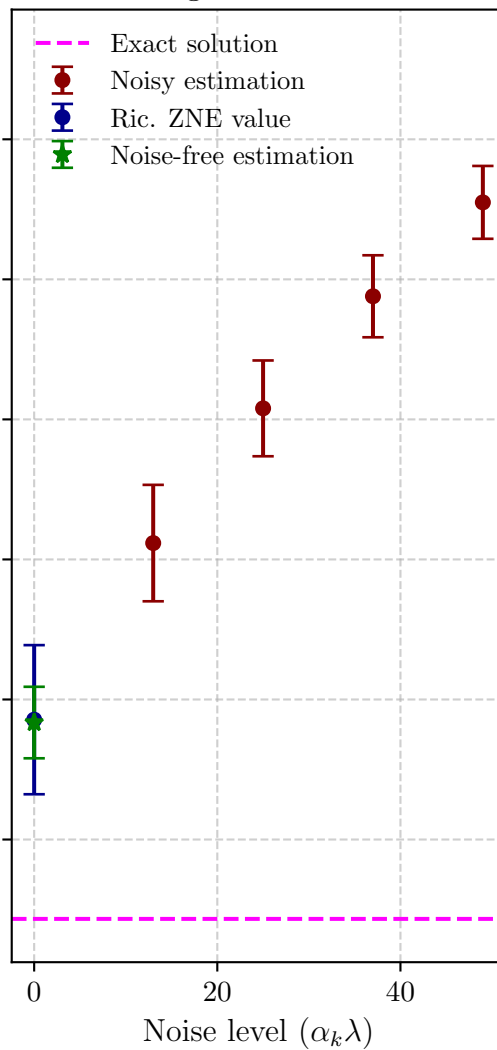
Heisenberg ansatz ric-3



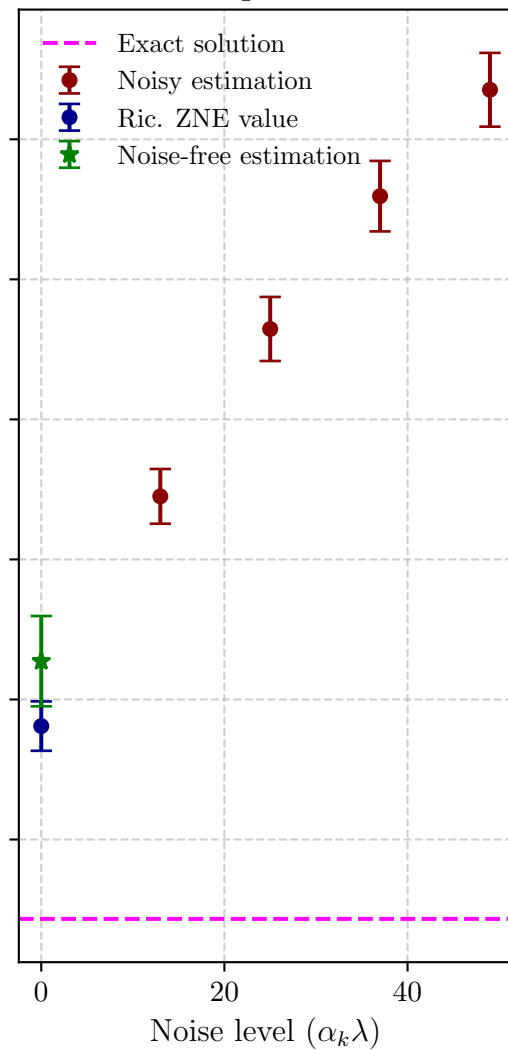
XY ansatz ric-4



Ising ansatz ric-4



Heisenberg ansatz ric-4



XY VQE CONFIGURATION

Parameter	Value
run	vqe
nqubits	7
state	dmatrix
output.file_name_prefix	xy_noisefree_time_evo
output.draw.status	True
output.draw.fig_dpi	100
output.draw.type	png
observable.def	ising
observable.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
observable.coefficients.bn	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
observable.coefficients.r	1
ansatz.layer	30
ansatz.gateset	1
ansatz.ugate.type	xy-iss
ansatz.ugate.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
ansatz.ugate.coefficients.bn	[0, 0, 0, 0, 0, 0, 0]
ansatz.ugate.coefficients.r	0
ansatz.ugate.time.min	0.0
ansatz.ugate.time.max	10.0
vqe.iteration	10
vqe.optimization.status	True
vqe.optimization.algorithm	SLSQP
vqe.optimization.constraint	True
init_param.value	random
noise_profile.status	True
noise_profile.type	dephasing
noise_profile.noise_prob	[0.001, 0.001, 0.0, 0.0]
noise_profile.noise_on_init_param.status	False
noise_profile.noise_on_init_param.value	0
redundant.identity_factors	[[0, 0, 0, 0], [1, 1, 1, 1]]
zne.method	richardson
zne.degree	1
zne.sampling	default
zne.data_points	

ISING VQE CONFIGURATION

Parameter	Value
run	vqe
nqubits	7
state	dmatrix
output.file_name_prefix	ising_noisefree_time_evo
output.draw.status	True
output.draw.fig_dpi	100
output.draw.type	png
observable.def	ising
observable.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
observable.coefficients.bn	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
observable.coefficients.r	1
ansatz.layer	30
ansatz.gateset	1
ansatz.ugate.type	ising
ansatz.ugate.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
ansatz.ugate.coefficients.bn	[0, 0, 0, 0, 0, 0, 0]
ansatz.ugate.coefficients.r	0
ansatz.ugate.time.min	0.0
ansatz.ugate.time.max	10.0
vqe.iteration	10
vqe.optimization.status	True
vqe.optimization.algorithm	SLSQP
vqe.optimization.constraint	True
init_param.value	random
noise_profile.status	True
noise_profile.type	dephasing
noise_profile.noise_prob	[0.001, 0.001, 0.0, 0.0]
noise_profile.noise_on_init_param.status	False
noise_profile.noise_on_init_param.value	0
redundant.identity_factors	[[0, 0, 0, 0], [1, 1, 1, 1]]
zne.method	richardson
zne.degree	1
zne.sampling	default
zne.data_points	

HEISENBERG VQE CONFIGURATION

Parameter	Value
run	vqe
nqubits	7
state	dmatrix
output.file_name_prefix	heisenberg_noise_free_time_evo
output.draw.status	True
output.draw.fig_dpi	100
output.draw.type	png
observable.def	ising
observable.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
observable.coefficients.bn	[1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0]
observable.coefficients.r	1
ansatz.layer	30
ansatz.gateset	1
ansatz.ugate.type	heisenberg
ansatz.ugate.coefficients.cn	[0.5, 0.5, 0.5, 0.5, 0.5, 0.5]
ansatz.ugate.coefficients.bn	[0, 0, 0, 0, 0, 0, 0]
ansatz.ugate.coefficients.r	0
ansatz.ugate.time.min	0.0
ansatz.ugate.time.max	10.0
vqe.iteration	10
vqe.optimization.status	True
vqe.optimization.algorithm	SLSQP
vqe.optimization.constraint	True
init_param.value	random
noise_profile.status	True
noise_profile.type	dephasing
noise_profile.noise_prob	[0.001, 0.001, 0.0, 0.0]
noise_profile.noise_on_init_param.status	False
noise_profile.noise_on_init_param.value	0
redundant.identity_factors	[[0, 0, 0, 0], [1, 1, 1, 1]]
zne.method	richardson
zne.degree	1
zne.sampling	default
zne.data_points	